Candidate's Number ……

THE PRESIDENT'S OFFICE
REGIONAL ADMINSTRATION AND LOCAL GOVERNMENT
IRINGA REGION FORM FOUR MOCK EXAMINATION
CHEMISTRY 1

CODE: 032/1

TIME: 3 Hours **23rd July, 2021**

INSTRUCTIONS

1. This paper consists of section A, B and C with a total of fourteen (14) questions.
2. Answer all questions in section A and B and only one (01) question from section C.
3. Cellular phones and unauthorized materials are NOT allowed in the examination room but non-programmable calculators are allowed.
4. Write your examination number on every page of your answer sheets provided
5. Remove this front page and attach together with your answer sheet
6. The following constants may be used:
   - Atomic masses H = 1, C = 12, O = 16, Na = 23, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Fe = 56, Cu = 64.
   - Avogadro's number = 6.02×10^{23}
   - GMV at STP = $22.4dm^3$
   - 1 Faraday = 96500 coulombs
   - 1 Litre = $1dm^3 = 1000cm^3$

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QUESTION NUMBER	**PUT (√) FOR THE QUESTIONS YOU ATTEMPT**	**SCORE AWARDED**	**INITIALS OF EXAMINER**
1			
2			
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13			
14			
TOTAL			

SECTION A: (15 Marks)

Answer all questions in this section.

1. For each of the items (i) – (x) choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet(s) provided.

i) Chemistry is all around us. This is because:……..

A. It gives us money when we learn to become doctors
B. Many of the materials we use in our daily lives involve chemistry.
C. It has many subject combination in high level of studies
D. It need laboratory rooms in most of its experiments.
E. It has many things to study which are also in other subjects.

ii) The following are examples of importances of first aid to a victim, except……

A. It develops hope and encouragement to a victim.
B. It gives employment to the first aid provider
C. It saves the life of a victim.
D. It serves the treatment costs
E. It reduces pain and suffering to a victim.

iii) In softening temporary hardness of water, too much slaked lime is not encouraged. This is because……

A. There will be replacement of temporary hardness and permanent hardness of water.
B. The whole water would turn alkaline because slaked lime is base
C. The slaked lime are much expensive, should be in controlled amount.
D. When removing hardness, it leaves limescale on the wall of the container
E. When present in excess, it will be difficult to be separated out after softening of water

iv) The oxide of the metal which is white when cold but yellow when hot is …….

A. Aluminium oxide B. Copper (II) oxide C. Zinc Oxide
D. Iron (III) oxide E. Iron (II) oxide

v) Why 96^0 C is referred as transitional temperature of sulphur?

A. It is the melting point of sulphur.
B. It is the boiling point of sulphur.
C. It is neither the boiling point nor the melting point of sulphur.
D. It interchanges rhombic sulphur and monoclinic sulphur.
E. It interchanges the α – sulphur and rhombic sulphur.

vi) The number of ions in a solution of 7.4g of $Ca(OH)_2$ is ……...

A. 6.02×10^{23} B. 1.204×10^{23} C. 6.02×10^{22}
D. 1.204×10^{22} E. 1.806×10^{23}

vii) The mixture of iodine and sand can be separated out by sublimation method through application of heat. Which one in the mixture can sublime?

A. Iodine B. Sand C. Heat
D. Both sand and Iodine E. Heat and Iodine

viii) How can you apply the knowledge of neutralization reaction to a person stung by wasp?

A. By rubbing with ash B. By removing the sting

C. By applying baking powder
D. By rubbing with cucumbers
E. By applying vinegar.

ix) Among the allotropes of carbon, one of them can be used as electrode during electrolysis because……..

A. It is inert electrode
B. It is active electrode.
C. It is slippery carbon.
D. It posses delocalized electrons.
E. It is called graphite.

x) The effect of catalyst on the rates of chemical reaction is …….

A. To lower the activation energy.
B. To speed-up the rates of reaction
C. To slow down the rate of reaction
D. To favour forward reaction in equibrium
E. To favour backward reaction in equilibrium.

2. Match the items in **List A** with responses in **List B** by writing the letter of the correct responses beside the item number in the answer sheet(s) provided.

List A	List B
i) Solute	A. It condenses the vapour of liquid which has not reached its boiling point
ii) Solvent	B. It loses its identity when in solution
iii) Air	C. Can be separated by simple distillation
iv) Fractionating column	D. It condenses the vapour of liquid which has reached its boiling point
v) Condenser	E. It consists of only pure colourless gas
	F. It is an apparatus in simple distillation
	G. Can be separated by fractional distillation
	H. The only best example is water
	I. It maintains its identity when in solution
	J. The good example is alcohol.

SECTION B: (70 Marks)

Answer all questions in this section

3. a) i) Some pieces of glass apparatus have the word PYREX written on them. What does this tell you?

ii) State any two strong reasons for the Bunsen burner to be the best heat source in the laboratory.

b) In tabular form, analyse any four (4) differences between boiling and evaporation.

4. a) Calculate the degree of saturation by 50g of potassium nitrate salt in 250cm^3 of water at room temperature, given that the density of water is 1g/cm^3.

b) In three points, state on how the chemical symbols of elements significant.

5. a) List down the uses of metal in reflection to the following physical properties in the table below.

Physical properties	Uses
i) Ductility	
ii) Sonority	
iii) Malleability	
iv) Lustrous	

b) Point out any three fields of application of scientific procedures.

6. a) Summarize the caution delivered by the warning signs with the names: Fragile, Oxidizing agent, Toxic and Flammable.

b) With example of reaction equation, show that lead (II) oxide is an amphoteric oxide.

7. a). Using balanced chemical equation show how;

i) Ethene undergoes additional reaction with hydrogen chloride.

ii) Ethanol reacts with Sulphuric acid at 1atm and 180^0 C.

b) Write down the structures of the following compounds

i) 2-methylpent-2-ene

ii) 4-methylpentan-2-ol

iii) Butanoic acid

8. a) Give a strong reason for the following statements:

i) A burning magnesium can not stop in a gas jar with carbondioxide gas.

ii) Metals are reducing agents.

b) Draw a well labelled diagram to show the steps of extraction of sulphur by Frasch process.

c) Analyse any two characteristics of a good fuel.

9. a) In acid base titration, the volume from the burette used was 25.00cm^3 of the monobasic mineral acid (HA) and dication base, $X(OH)_2$ with the same molar concentrations. What is the volume capacity of the pipette used?

b) What is fire triangle? In brief, summarize the process of stopping fire.

10. a) What is the difference between hard water and soft water?

b) When metal X is introduced into soft water, dissolved with evolution of a gas, it turned into milky coloured, when left near the burning ethane gas.

i) Identify the metal X and then calculate its number of neutrons

ii) Write a net ionic equation by showing the necessary steps for the dissolution of X.

iii) Explain the formation of milky coloured mixture by supporting with reaction equations starting with the burning of ethane gas (Hint: All equations must be balanced)

11. Contact process is defined by the equilibrium reaction equation. $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$

a) Define the term equilibrium reaction in terms of the rates of reaction and sketch the graph to present the reaction equations for the forward, backward and equilibrium state.

b) If the reaction is in a closed system, which principle governs the position of reactions in a system? Give its statement.

12. a) How the following substances can be identified in the laboratory?
    i) Hydrogen gas
    ii) Oxygen gas
    iii) Hydrogen chloride gas
    iv) A nitrate compounds.

    b) Explain how salt crystals can be separated out from the mixture with pieces of glass (3 steps only)

SECTION C (15 Marks)

Answer one (1) question from this section

13. a) In electrolytic production of hydrogen gas dilute mineral acids is used. Which method is used in its collection? Give a reason.

    b) If 0.2 Faraday is used in production of hydrogen gas electrolytically, then the gas formed allowed to pass through a hot tube with a sample of copper (II) oxide and copper (II) sulphate in which chemical changes and physical changes occurred in the sample.
        i) Which compound between copper (II)Oxide and copper (II) sulphate encountered chemical change and physical change?

        ii) If the product for the chemical changes involved, one of its product was 10g of the solid and other product was liquid. Calculate the volume of the hydrogen gas that should be supplied and the volume of the hydrogen gas remained respective to 0.2 Faraday at S.T.P.

14. a) Analyse the deficiency outcomes for nitrogenous fertilizer called NPK to the plant respective to principal mineral content.

    b) Describe any four water pollutants.